

(bottom) side, a left and right side, and an anterior (front) and posterior (back) end. The cell movements that begin gastrulation occur on the dorsal side, opposite where the sperm entered the egg. The frog's anus develops from the blastopore, and the mouth eventually breaks through at the opposite end of the archenteron.

Gastrulation in Chicks

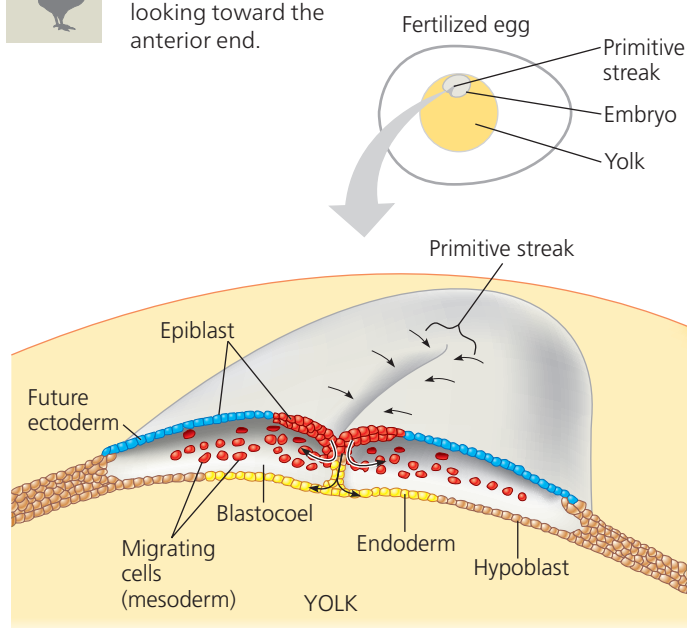
At the onset of gastrulation in chicks, an upper and a lower layer of cells—the *epiblast* and *hypoblast*—lie atop a yolk mass. All the cells that will form the embryo come from the epiblast. During gastrulation, some epiblast cells move toward the midline, detach, and move inward toward the yolk (Figure 47.11). The pileup of cells moving inward at the midline produces a visible thickening called the *primitive streak*. Some of these cells move downward and form endoderm, pushing aside the hypoblast cells, while others migrate laterally (sideways) and form mesoderm. The cells left behind on the surface at the end of gastrulation will become ectoderm. The hypoblast cells later segregate from the endoderm and eventually form part of the sac that surrounds the yolk and also part of the stalk that connects the yolk mass to the embryo.

Although different terms describe gastrulation in different vertebrate species, the rearrangements and movements of cells exhibit a number of fundamental similarities. In particular, the primitive streak, shown in Figure 47.11 for the chick embryo, is the counterpart of the blastopore lip, shown in Figure 47.10 for the frog embryo. Formation of a primitive streak is also central to human embryo gastrulation, our next topic.



▼ **Figure 47.11 Gastrulation in a chick embryo.**

This is a cross section of a gastrulating embryo, looking toward the anterior end.



Gastrulation in Humans

Unlike the large, yolky eggs of many vertebrates, human eggs are quite small, storing little in the way of food reserves. Fertilization takes place in the oviduct, and development begins while the embryo completes its journey down the oviduct to the uterus (see Figure 46.15).

Figure 47.12 outlines the development of the human embryo, starting about 6 days after fertilization. This depiction is largely based on observations of embryos from other mammals, such as the mouse, and of very early human embryos following *in vitro* fertilization.

- 1 At the end of cleavage, the embryo has more than 100 cells arranged around a central cavity and has reached the uterus. At this stage, the embryo is called a **blastocyst**, the mammalian version of a blastula. Clustered at one end of the blastocyst cavity is a group of cells called the **inner cell mass**, which will develop into the embryo proper. It is the cells of the inner cell mass that are the source of embryonic stem cell lines (see Concept 20.3).
- 2 Implantation of the embryo is initiated by the **trophoblast**, the outer epithelium of the blastocyst. Enzymes secreted by the trophoblast during implantation break down molecules of the endometrium, the lining of the uterus, allowing invasion by the blastocyst. The trophoblast also extends finger-like projections that cause capillaries in the endometrium to spill out blood that can be captured by trophoblast tissues. Around the time the embryo undergoes implantation, the inner cell mass of the blastocyst forms a flat disk with an inner layer of cells, the *epiblast*, and an outer layer, the *hypoblast*. As is true for a bird embryo, the human embryo develops almost entirely from epiblast cells.
- 3 Following implantation, the trophoblast continues to expand into the endometrium, and four new membranes appear. Although these **extraembryonic membranes** arise from the embryo, they enclose specialized structures located outside the embryo. As implantation is completed, gastrulation begins. Some epiblast cells remain as ectoderm on the surface, while others move inward through a primitive streak and form mesoderm and endoderm, just as in the chick (see Figure 47.11).
- 4 By the end of gastrulation, the embryonic germ layers have formed. Extraembryonic mesoderm and four distinct extraembryonic membranes now surround the embryo. As development proceeds, cells of the invading trophoblast, the epiblast, and the adjacent endometrial tissue all contribute to the formation of the placenta. This vital organ mediates the exchange of nutrients, gases, and nitrogenous wastes between the developing embryo and the mother (see Figure 46.16).

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